

牛顿环

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1 实验数据

环数:k	10	15	20	25	30	35	40	45
左读数: x_1/mm	29.609	30.149	30.698	31.179	31.622	32.021	32.390	32.731
右读数: x_2/mm	23.931	23.285	22.765	22.288	21.856	21.480	21.085	20.750
差值: $(x_1 - x_2)/mm$	5.678	6.864	7.933	8.891	9.766	10.541	11.3	11.981
差值平方: $(x_1 - x_2)^2/mm^2$	32.24	47.11	62.93	79.05	95.37	111.11	127.8	143.54

2 最小二乘法

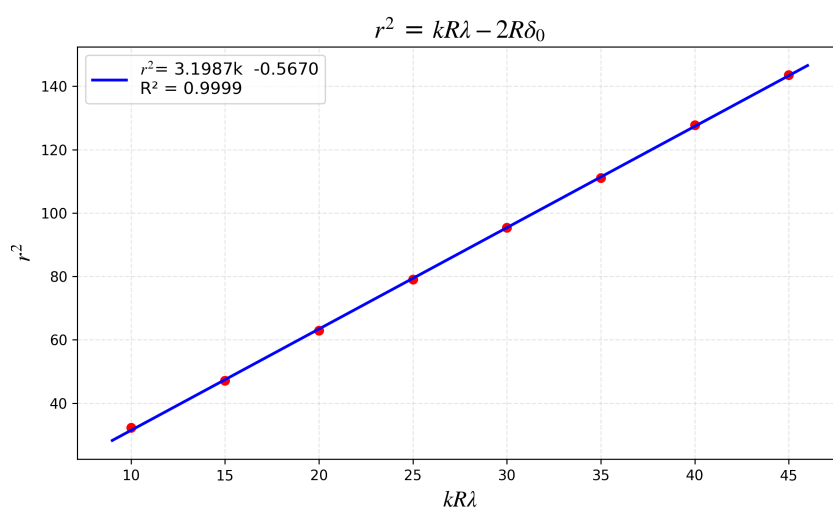


图 1: 最小二乘法

注意, 这里的 r^2 指的是圆环直径的平方, 所以要 $r^2 = 4R_{\text{圆环}}^2$

$$\begin{cases} \text{斜率} : 3.198668654761905 \\ \text{截距} : -0.5669663809523939 \\ R^2 \text{值} : 0.9998704236040188 \end{cases}$$

我们知道了斜率 $k_1 = 3.198668654761905$, 则

$$R = \frac{k_1}{4 \times \lambda} \times 10^{-6} = \frac{3.198668654761905}{589.3 \times 10^{-9} \times 4} \times 10^{-6} m = 1.3569780480069173 m$$

3 逐差法

$$Y_1 = D_{45}^2 - D_{25}^2 = 64.49448000000007mm^2$$

$$Y_2 = D_{40}^2 - D_{20}^2 = 64.87053599999999mm^2$$

$$Y_3 = D_{35}^2 - D_{15}^2 = 63.998185mm^2$$

$$Y_4 = D_{30}^2 - D_{10}^2 = 63.13507199999995mm^2$$

$$R_1 = \frac{Y_1}{\lambda \times 20 \times 4} = 1.3680315628712043m$$

$$R_2 = \frac{Y_2}{\lambda \times 20 \times 4} = 1.3760083149499405m$$

$$R_3 = \frac{Y_3}{\lambda \times 20 \times 4} = 1.3575043483794331m$$

$$R_4 = \frac{Y_4}{\lambda \times 20 \times 4} = 1.3391963346343108m$$

$$R = \frac{R_1 + R_2 + R_3 + R_4}{4} = 1.3601851402087222m$$

4 误差分析

4.1 最小二乘法的误差分析

由于我们只进行了一次的测量, 所以没有 A 类不确定度.
我们的螺旋测微器的极限为 $0.01mm$, 则

$$\Delta_R = 2.449305401279594 \times 10^{-3}m$$

则

$$R = (1.3569780480069173 \pm 2.449305401279594 \times 10^{-3})m$$

4.2 逐差法的误差分析

$$\mu(D_k) = \frac{1 \times 10^{-5}}{\sqrt{3}}m = 5.7735 \times 10^{-6}m$$

$$\mu(R) = \sqrt{\sum_k \left(\frac{\partial R}{\partial D_k} \times \mu(D_k) \right)^2} = \frac{\mu(D_k)}{160\lambda} \sqrt{\sum_k D_k^2} = 1.6191033280672995 \times 10^{-3}m$$

则

$$R = \frac{R_1 + R_2 + R_3 + R_4}{4} = (1.3601851402087222 \pm 1.6191033280672995 \times 10^{-3})m$$